

Gradient-Driven Criticality: a MOND Sector That Cures Its Own Gradient Instability, and a Marginal State That Is Exactly Cold

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Abstract

A relativistic MOND candidate built from a cusciton clock τ , a dynamical scalar χ with a projected-gradient sector $sW(Y, \tau)$ and a cubic coupling has a sub-horizon sound speed $c_s^2 = (1 - s_0)m_{\text{rel}}/(2 - m_{\text{rel}})$, negative whenever the clock rate $s_0 = \dot{\tau}$ exceeds unity [3]. That result was obtained on a background with *no* field gradient. Since the sector’s entire content is its dependence on the projected gradient invariant $Y = |\nabla_\perp \chi|^2$ — the function $W(Y)$ is the interpolating function — we recompute the sound speed with a background gradient carried through. Three consequences follow from the candidate’s own closure with nothing added: the destabilising stiffness falls, $W_Y = d(1 + Y/\ell)^{-1/2} \rightarrow 0$; the logarithm margin *grows*, $m(Y) = m_0 + 2dY$, carrying the background away from its singularity; and c_s^2 rises monotonically and crosses zero at a unique gradient Y_* . Below Y_* the sector is unstable so gradients grow and Y rises; above it the sector is stable so expansion dilutes Y and it falls. Y_* is a two-sided attractor, and the state it selects has $c_s^2 = 0$ exactly. Evaluated on the candidate’s own frozen coefficient history, Y_* lies at 0.2–2% of the transition scale, so an infinitesimal gradient suffices, and at $Y = \ell$ the sound speed stays positive even with the clock rate tripled. We then show that this surface is not merely dynamical: varying the action with respect to the inverse metric gives $\det(T_{tx}^{\text{mixed}} - L\mathbb{I}) = -s_0 b^2 W \mathcal{N}$, where $\mathcal{N}$ is exactly the numerator whose vanishing defines Y_* , so on the critical surface a third eigenvalue of the mixed stress joins the transverse pair and the sector’s stress takes perfect-fluid form. Isotropic stress together with non-propagating perturbations is precisely a clustering cold component — the ingredient a Lean-certified necessity result requires of any completion — and both properties arrive together on one surface fixed by the action rather than by tuning. Seven statements are machine-checked in Lean 4. What is not established is the tracking dynamics: the Lyman- α bound $c_s^2 \leq 10^{-9}$ requires the gradient to follow Y_* to a fractional precision of 3×10^{-7} to 2×10^{-6} , and whether the back-reaction achieves that is the next calculation.

1 The instability, and why its derivation was incomplete

The repository [1] studies a MOND kernel on the de Sitter-locked acceleration scale $a_0 = \kappa c \sqrt{G\rho_\Lambda}$, and its Lean-certified necessity result states that any completion passing galaxies, clusters, the CMB and the Lyman- α forest together must contain a real, clustering cold component. The programme’s lead candidate [2] is, in signature $(-+++)$ with $c = 1$,

$$S = \int d^4x \sqrt{-g} \left[\frac{M^2}{2} (R - 2\Lambda) + P(X, \tau) - V(\tau) + sW(Y, \tau) + \gamma X \square \chi \right] + S_r + S_b, \quad (1)$$

with $s = \sqrt{-\nabla_\mu \tau \nabla^\mu \tau}$, $n_\mu = -\partial_\mu \tau / s$, $Q = n^\mu \partial_\mu \chi$, $Y = (g^{\mu\nu} + n^\mu n^\nu) \partial_\mu \chi \partial_\nu \chi$ and $X = Q^2 - Y$. The clock is linear in s , hence a cuscutoon [4], and its field equation is a constraint. Its coefficient closure, on the branch studied there, reads $W = U$, $W_Y = d$, $P_X = Ud/m$, $P_{XX} = 2Ud^2/m^2$ with $m = U - 2dq^2$ the logarithm margin and $m_{\text{rel}} = m/U$.

The clock stability theorem [3] eliminates the constraint and the lapse in the sub-horizon limit and obtains

$$c_s^2 = \frac{2P_X(1-D) - 2s_0 W_Y}{B(1-D)}, \quad B = 2P_X + 4XP_{XX}, \quad D = \frac{2Q^2 W_Y}{W}, \quad (2)$$

which under the closure collapses to $c_s^2 = (1-s_0)m_{\text{rel}}/(2-m_{\text{rel}})$: the sector is gradient-stable if and only if the clock does not run faster than proper time. On the candidate's own history s_0 crosses unity and the late branch is unstable, which was recorded as a cost of the candidate.

Equation (2), however, was evaluated with W_Y and W taken at $Y = 0$. That is a background with no field gradient anywhere — and the entire MOND content of (1) sits in the Y -dependence of W . The question this note answers is what the same reduction gives when the background has a gradient.

2 The sound speed at finite gradient

Work in the WKB limit with a locally uniform background gradient. The constraint structure of [3] carries over unchanged; only the coefficients are now evaluated at the background Y , and the stiffness that appears depends on the polarisation,

$$\mathcal{W} = \begin{cases} W_Y(Y) & k \perp \nabla \chi, \\ W_Y(Y) + 2YW_{YY}(Y) & k \parallel \nabla \chi, \end{cases} \quad c_s^2(Y) = \frac{2P_X(1-D) - 2s_0 \mathcal{W}}{B(1-D)}, \quad D = \frac{2Q^2 \mathcal{W}}{W(Y)}. \quad (3)$$

Three things now follow from the closure, none of them put in by hand. With $W(Y) = U + 2d\ell(\sqrt{1+Y/\ell} - 1)$:

$$W_Y = \frac{d}{\sqrt{1+Y/\ell}}, \quad W_Y + 2YW_{YY} = \frac{d}{(1+Y/\ell)^{3/2}}, \quad m(Y) = U - 2dX = m_0 + 2dY. \quad (4)$$

The destabilising term $-2s_0 \mathcal{W}$ therefore *shrinks* with the gradient in either polarisation; D shrinks with it; and the logarithm margin *grows*, carrying $P_X = Ud/m(Y)$ away from the singularity that the closure's logarithm sits against. Hence $c_s^2(Y)$ increases monotonically from $(1-s_0)m_{\text{rel}}/(2-m_{\text{rel}})$ and, whenever that starting value is negative, crosses zero at a unique Y_* .

3 The attractor

Below Y_* the sector is gradient-unstable, so perturbations grow, so the field gradient grows, so Y rises. Above Y_* the sector is stable, nothing pumps the gradient, and expansion dilutes it, so Y falls. Y_* is approached from both sides. The state it selects has $c_s^2 = 0$: the sector's perturbations do not propagate at all.

Table 1 evaluates (3) on the candidate's own frozen coefficient functions and its archived backward branch, imported read-only. The critical gradient lies far *below* the transition scale, at 0.2–2% of ℓ and 4–15% in gradient units, so only an infinitesimal field gradient is needed: the marginal state

Table 1: The critical gradient on the candidate’s own history (transverse polarisation).

a	s_0	$c_s^2(Y=0)$	Y_*/ℓ	$c_s^2(Y_*/2)$	$c_s^2(2Y_*)$
1.00	1.032	-1.51×10^{-3}	0.0061	-7.4×10^{-4}	$+1.4 \times 10^{-3}$
0.75	1.072	-2.57×10^{-3}	0.0104	-1.2×10^{-3}	$+2.2 \times 10^{-3}$
0.56	1.173	-4.29×10^{-3}	0.0174	-2.0×10^{-3}	$+3.2 \times 10^{-3}$
0.42	1.413	-5.24×10^{-3}	0.0213		
0.32	1.118	-4.86×10^{-4}	0.0019		

is easy to reach, in voids as much as in halos. The healthy-kinetic domain extends to 42–484 ℓ , so Y_* sits well inside it. And at $Y = \ell$ the sound speed remains positive on every unstable epoch even when s_0 is tripled, whereas the same tripling at $Y = 0$ makes it negative: above Y_* the gradient decides the sign and the clock only rescales the magnitude.

4 What the critical surface is

The surface has an algebraic meaning that owes nothing to the WKB argument. Write $\mathcal{N} = 2P_X(1 - 2Q^2W_Y/W) - 2s_0W_Y$, the numerator of (3), and $L = P - V + s_0W$. Varying the density of (1) with respect to the inverse metric in all ten components gives, on a background carrying gradient $b = \sqrt{Y}$ in one direction,

$$T_{00} = 2P_XQ^2 - P + V, \quad T_{11} = L + 2(P_X - s_0W_Y)b^2, \quad T_{22} = T_{33} = L, \quad T_{01} = 2P_XQb, \quad (5)$$

so two eigenvalues of the mixed tensor $T^\mu{}_\nu$ equal L for any background. The remaining block satisfies the exact identity

$$\det(T_{tx}^{\text{mixed}} - L\mathbb{I}) = -s_0b^2W\mathcal{N} \quad (6)$$

This identity was recorded in a concurrent independent audit of the same action and is verified here with an independent implementation. Its consequence is that $\mathcal{N} = 0$ — the same condition that defines Y_* — is precisely the condition for a *third* eigenvalue to join the transverse pair. On the critical surface the mixed stress takes perfect-fluid form and the sector’s anisotropic stress vanishes. The degeneracy is non-trivial: the shifted determinant carries an overall b^2 , so a gradient-free background is degenerate for an unrelated reason, and it is finite gradient that makes $\mathcal{N} = 0$ meaningful.

So the marginal state has isotropic stress and perturbations that do not propagate. Those two properties together are what a clustering cold component is, and here they arrive on one surface fixed by the action rather than being imposed as separate conditions on a fluid.

5 Machine-checked statements

The following are theorems of the repository’s Lean 4 file (`Mondlean.lean`, 111 theorems, no sorry; each depends only on `propext`, `Classical.choice`, `Quot.sound`). `mond_stiffness_strict_anti`: for $d, \ell > 0$ the destabilising stiffness $d/\sqrt{1 + Y/\ell}$ is strictly decreasing in Y . `mond_margin_strict_mono`: the margin $m_0 + 2dY$ is strictly increasing. `critical_gradient_unique`: a continuous strictly monotone quantity negative at one gradient and

positive at another has exactly one marginal point between them. `stress_degeneracy_identity`: identity (6), written out in the block entries. `critical_stiffness_kills_N`: the critical stiffness, stated as $W_Y(2P_X Q^2 + s_0 W) = P_X W$, makes $\mathcal{N}$ vanish. Two further theorems from [3] supply the $Y = 0$ closure and the stability equivalence. What Lean certifies here is algebra and logic; the evaluations in Table 1 are numerical and are labelled as such.

6 What is not established

The sign structure is established; the dynamics are not. The sound speed rises steeply out of the marginal point, so the Lyman- α requirement $c_s^2 \leq 10^{-9}$ demands that the gradient track Y_* to a fractional precision of 3×10^{-7} to 2×10^{-6} . Criticality therefore replaces a tuning of the logarithm margin to 10^{-8} with a dynamical tracking requirement of comparable sharpness, and whether the back-reaction achieves it is an integration that has not been done. Nor does the candidate’s particular coefficient history pass the forest: its early, stable branch sits at $c_s^2 \approx 10^{-4}$, because criticality operates only where the clock runs fast. The honest form of the new requirement is that a history whose clock runs faster than proper time at all epochs of interest is driven to exact isotropy and zero sound speed by its own MOND sector — a condition on the coefficient functions that can be imposed a priori rather than fitted. No claim is made here about the amount of the sector, the acoustic peaks, lensing, or the post-Newtonian limit.

7 Reproducibility

`fable_independent_2026/L192_gradient_criticality.py` and `L193_stress_degeneracy.py` with their committed outputs (commits `5fdec3d9a`, `bc71158f0`); the Lean file `fable_independent_2026/lean_2026/Mondlean.lean`. The candidate’s coefficient functions and archived branch are imported read-only from the collaborating programme’s directory named in [2].

References

- [1] The research repository, <https://github.com/carlzimmerman/zimmerman-formula> (2026).
- [2] C. P. Zimmerman, *A Primordial-Clock Relativistic MOND Candidate: Status Report*, doi:10.5281/zenodo.22699828 (2026).
- [3] C. P. Zimmerman, *The Clock Stability Theorem*, doi:10.5281/zenodo.22717950 (2026).
- [4] N. Afshordi, D. J. H. Chung, G. Geshnizjani, Phys. Rev. D **75**, 083513 (2007).
- [5] M. Milgrom, Astrophys. J. **270**, 365 (1983).